GloVe: The Theory

Why ratios of co-occurrence probabilities lead to a log-bilinear model

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Global Vectors for Word Representation — Pennington, Socher & Manning (2014)

October 13, 2026

Roadmap

1. Representing meaning with vectors
2. Co-occurrence statistics
3. Deriving the GloVe (Global Vectors) model
4. The weighted objective
5. Connections: pointwise mutual information (PMI), singular value decomposition (SVD), skip-gram
6. Complexity
7. Empirical behaviour & limitations

Abbreviations used in this deck

AdaGrad	Adaptive gradient (optimiser)	nnz	Number of non-zero entries
BERT	Bidirectional Encoder Representations from Transformers	PMI	Pointwise mutual information
		PPMI	Positive PMI (negatives set to 0)
ELMo	Embeddings from Language Models	SG	Skip-gram (word2vec model)
GloVe	Global Vectors for Word Representation	SVD	Singular value decomposition
LSA	Latent Semantic Analysis		

Venue abbreviations (EMNLP, NeurIPS, JMLR, JASIS) are spelled out in the References.

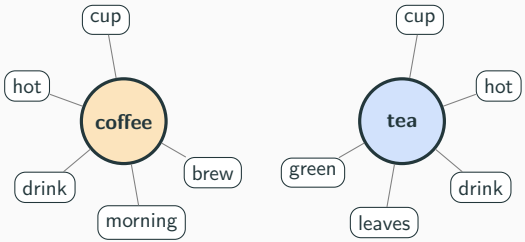
Representing meaning

The distributional hypothesis

"You shall know a word by the company it keeps."

— J. R. Firth, 1957

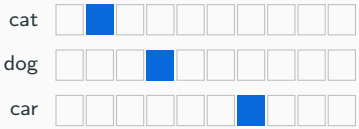
- Words used in similar contexts tend to have similar meanings.
- So a word's meaning can be estimated from statistics of the words around it.
- Every embedding method — Latent Semantic Analysis (LSA), word2vec, GloVe — builds on this idea.



shared contexts ⇒ similar vectors

From symbols to geometry

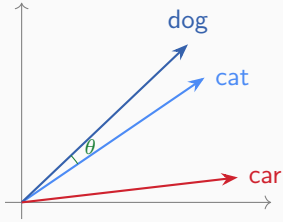
One-hot ($\mathbb{R}^V$, sparse)



all pairs orthogonal:

$\cos(\text{cat}, \text{dog}) = \cos(\text{cat}, \text{car}) = 0$

Dense embedding ($\mathbb{R}^d$, $d \ll V$)



$$\text{similarity} = \cos\theta = \frac{\mathbf{w}_a^\top \mathbf{w}_b}{\|\mathbf{w}_a\|\|\mathbf{w}_b\|}$$

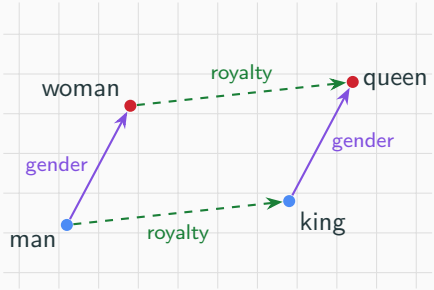
The goal: linear structure

Good embeddings encode relations as consistent directions:

$$\mathbf{w}_{\text{king}} - \mathbf{w}_{\text{man}} + \mathbf{w}_{\text{woman}} \approx \mathbf{w}_{\text{queen}}$$

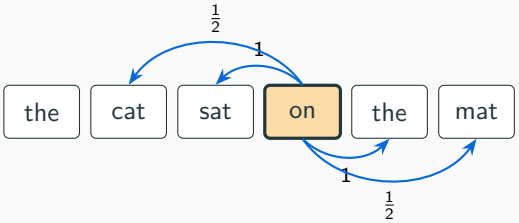
$$\mathbf{w}_{\text{Paris}} - \mathbf{w}_{\text{France}} + \mathbf{w}_{\text{Italy}} \approx \mathbf{w}_{\text{Rome}}$$

This is a *design target*. The question GloVe asks is:
what model of the statistics makes such directions appear?



Co-occurrence statistics

The co-occurrence matrix X

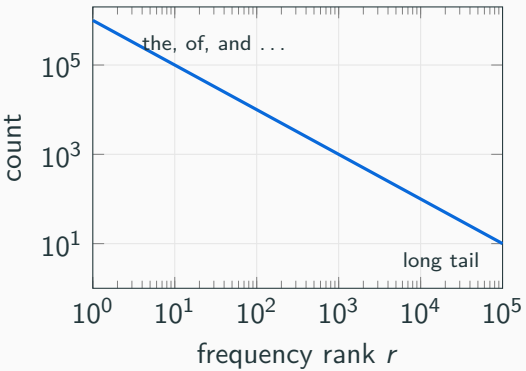


X_{ij} = how often j appears within a window around i. Closer words count more (weight 1/distance).

	the	cat	mat	on	sat
the		1	1	1	1
cat	1			0.5	1
mat	1			0.5	
on	1	0.5	0.5		1
sat	1	1		1	

symmetric, mostly zeros, and global: it summarises the whole corpus

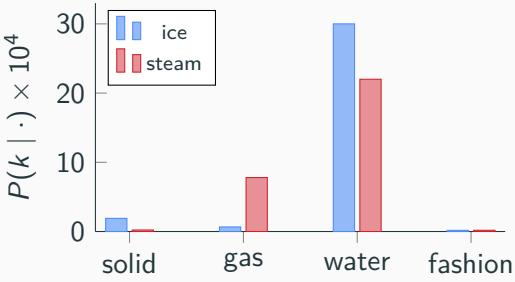
Counts are heavy-tailed (Zipf)



- Word frequency $\propto 1/\text{rank}$, so a handful of words dominate.
- Co-occurrence counts are also heavy-tailed: a few huge cells and many cells equal to 1.
- Consequences for any model:
  - raw counts are dominated by stop words;
  - small counts are noisy;
  - $\log X$ compresses the range, but $\log 0 = -\infty$.

From counts to probabilities

$$X_i = \sum_k X_{ik}, \qquad P_{ik} = P(k \mid i) = \frac{X_{ik}}{X_i}$$



- Raw conditional probabilities are hard to compare: "water" is large for *both* words.
- What tells us about meaning is how the two distributions differ.

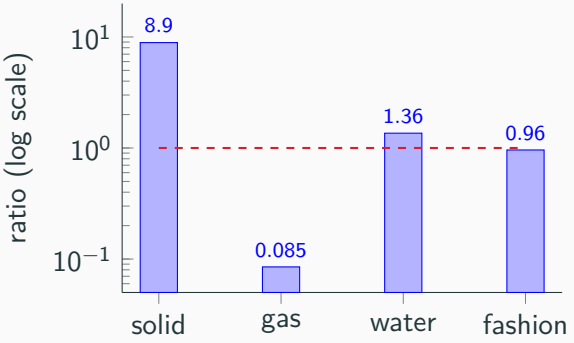
Values from Pennington et al. (2014), Table 1.

The key insight: ratios

$$\frac{P_{ik}}{P_{jk}} = \frac{P(k \mid \text{ice})}{P(k \mid \text{steam})}$$

k	ratio	meaning
solid	8.9	k relates to ice
gas	0.085	k relates to steam
water	1.36	relates to both
fashion	0.96	relates to neither

Ratios cancel out the probe word's own frequency and keep only what discriminates i from j.



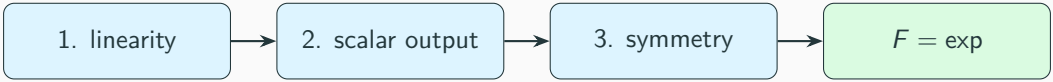
Deriving the model

What we want

Find vectors and a function F such that

$$F(\mathbf{w}_i, \mathbf{w}_j, \tilde{\mathbf{w}}_k) = \frac{P_{ik}}{P_{jk}}$$

- $\mathbf{w}_i, \mathbf{w}_j \in \mathbb{R}^d$ are *word* vectors; $\tilde{\mathbf{w}}_k$ is a separate *context* vector.
- The right-hand side is measured from the corpus; the left-hand side is the model.
- F could be anything at this point. We now add three natural constraints that pin it down.



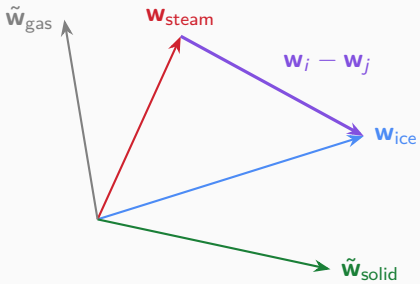
Constraints 1 & 2: differences and dot products

Linearity. Vector spaces are linear, so compare i and j through their difference:

$$F(\mathbf{w}_i - \mathbf{w}_j,\ \tilde{\mathbf{w}}_k)$$

Scalar output. The target is a scalar but the arguments are vectors. Use a dot product so that dimensions do not mix in arbitrary ways:

$$F\big((\mathbf{w}_i - \mathbf{w}_j)^\top \tilde{\mathbf{w}}_k\big) = \frac{P_{ik}}{P_{jk}}$$



$(\mathbf{w}_i - \mathbf{w}_j)^\top \tilde{\mathbf{w}}_k$ is large for solid and negative for gas, so it plays the role of log ratio

Constraint 3: symmetry forces $F = \exp$

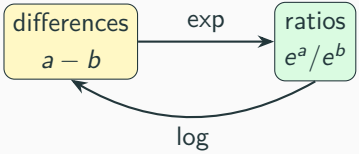
The roles of word and context are arbitrary: swapping $\mathbf{w} \leftrightarrow \tilde{\mathbf{w}}$ and $X \leftrightarrow X^\top$ should leave the model unchanged. Require F to be a homomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}_{>0}, \times)$:

$$F\big((\mathbf{w}_i - \mathbf{w}_j)^\top \tilde{\mathbf{w}}_k\big) = \frac{F(\mathbf{w}_i^\top \tilde{\mathbf{w}}_k)}{F(\mathbf{w}_j^\top \tilde{\mathbf{w}}_k)} \quad \Longrightarrow \quad F = \exp$$

Matching numerator to numerator:

$$\exp(\mathbf{w}_i^\top \tilde{\mathbf{w}}_k) = P_{ik} = \frac{X_{ik}}{X_i}$$

$$\mathbf{w}_i^\top \tilde{\mathbf{w}}_k = \log X_{ik} - \log X_i$$



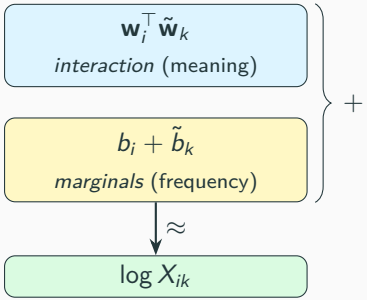
Restoring symmetry with biases

$\log X_i$ does not depend on k: absorb it into a bias b_i, and add $\tilde{b}_k$ for symmetry:

$$\mathbf{w}_i^\top \tilde{\mathbf{w}}_k + b_i + \tilde{b}_k = \log X_{ik}$$

- A log-bilinear model: bilinear in $(\mathbf{w}, \tilde{\mathbf{w}})$ and linear in $\log X$.
- The biases absorb word frequency, so the vectors are free to encode *relations*.
- Vector differences now reproduce log ratios:

$$(\mathbf{w}_i - \mathbf{w}_j)^\top \tilde{\mathbf{w}}_k = \log \frac{P_{ik}}{P_{jk}} + \text{const}$$



The weighted objective

Fitting it: why *weighted* least squares?

The equation cannot hold exactly, so fit it by regression. Unweighted least squares has three problems:

$\log 0$	**noise**	**dominance**
Most cells are 0 and $\log 0 = -\infty$.	Cells with count 1–2 are unreliable, yet they are the majority.	Very frequent pairs (*the*, *of*) would drive the fit.

$$J = \sum_{i,j=1}^{V} f(X_{ij}) \left(\mathbf{w}_i^\top \tilde{\mathbf{w}}_j + b_i + \tilde{b}_j - \log X_{ij} \right)^2$$

A single weighting function f fixes all three.

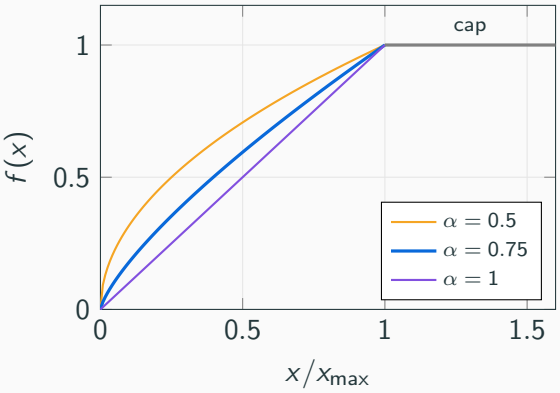
The weighting function

$$f(x) = \begin{cases} (x/x_{\max})^{\alpha} & x < x_{\max} \\ 1 & \text{otherwise} \end{cases}$$

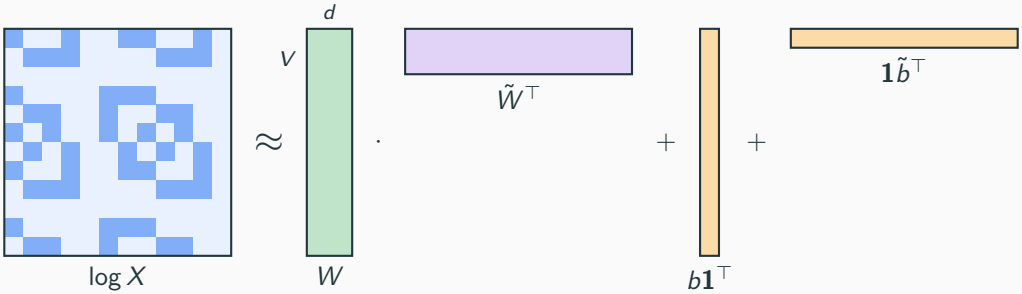
Required properties:

1. $f(0) = 0$, and $f(x)\log^2 x \to 0$ as $x \to 0$, so zero cells drop out and only the $\mathrm{nnz}(X)$ (number of non-zero) terms remain.
2. f is non-decreasing, so rare pairs are not over-weighted.
3. f is bounded, so frequent pairs are not over-weighted.

Paper: $\alpha = \frac{3}{4}$, $x_{\max} = 100$.



GloVe as weighted matrix factorisation



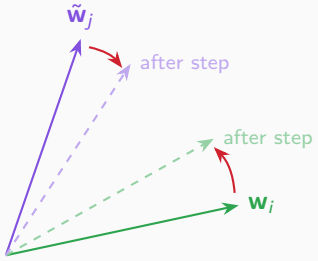
- A rank-d approximation of $\log X$, plus row and column offsets.
- The loss is evaluated only on observed cells, each weighted by $f(X_{ij})$, unlike SVD, which weights all cells equally.
- The problem is non-convex but well behaved, and is solved by stochastic gradient methods (AdaGrad, adaptive gradient, in the paper).

Learning dynamics: what a gradient step does

Let $e_{ij} = \mathbf{w}_i^\top \tilde{\mathbf{w}}_j + b_i + \tilde{b}_j - \log X_{ij}$ (prediction error).

$$\nabla_{\mathbf{w}_i} J = 2f(X_{ij})\, e_{ij}\, \tilde{\mathbf{w}}_j$$
$$\nabla_{\tilde{\mathbf{w}}_j} J = 2f(X_{ij})\, e_{ij}\, \mathbf{w}_i$$
$$\partial_{b_i} J = \partial_{\tilde{b}_j} J = 2f(X_{ij})\, e_{ij}$$

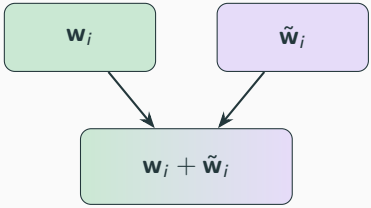
- $e < 0$ (pair seen *more* than predicted): pull $\mathbf{w}_i$ and $\tilde{\mathbf{w}}_j$ together.
- $e > 0$: push them apart.
- The step is scaled by f, so trusted (frequent) pairs move the vectors more.



$e_{ij} < 0$: increase $\mathbf{w}_i^\top \tilde{\mathbf{w}}_j$

Two vector sets, one embedding

- The model learns W (words) and $\tilde{W}$ (contexts).
- When X is symmetric, W and $\tilde{W}$ are equivalent: they differ only because of random initialisation.
- The paper uses $\mathbf{w}_i + \tilde{\mathbf{w}}_i$ as the final representation:
  - a cheap ensemble that reduces overfitting and noise;
  - a small but consistent gain on analogy tasks.
- Both vectors also enter similarity as $\mathbf{w}_i^\top \tilde{\mathbf{w}}_j + \ldots$, so the sum mixes first- and second-order similarity.



final word vector

Connections

GloVe and pointwise mutual information

$$\mathrm{PMI}(i,j) = \log \frac{P(i,j)}{P(i)P(j)} = \log X_{ij} - \log X_i - \log X_j + \log |X|$$

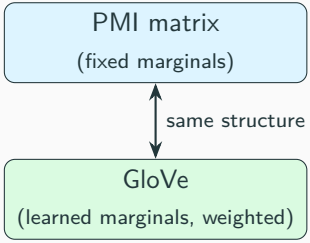
Compare with GloVe's model:

$$\mathbf{w}_i^\top \tilde{\mathbf{w}}_j = \log X_{ij} - b_i - \tilde{b}_j$$

If the learned biases were $b_i = \log X_i$ and $\tilde{b}_j = \log X_j - \log |X|$, then

$$\mathbf{w}_i^\top \tilde{\mathbf{w}}_j = \mathrm{PMI}(i,j).$$

GloVe can be read as a weighted low-rank factorisation of a (shifted) PMI matrix, where the shift is learned through the free biases instead of being fixed.



GloVe and skip-gram (word2vec)

Skip-gram (SG) models $Q_{ij} = \dfrac{\exp(\mathbf{w}_i^\top \tilde{\mathbf{w}}_j)}{\sum_k \exp(\mathbf{w}_i^\top \tilde{\mathbf{w}}_k)}$. Summed over the corpus, its loss groups into

$$J_{\mathrm{SG}} = -\sum_{i,j} X_{ij} \log Q_{ij} = \sum_i X_i \, H(P_i, Q_i)$$

a *frequency-weighted cross-entropy* between the empirical P_i and the model Q_i.

Problems

- Normalising Q costs $O(V)$ per term.
- Cross-entropy puts too much weight on unlikely events.
- The weight X_i is unbounded.

GloVe's replacements

- Unnormalised $\hat{P}_{ij} = X_{ij}$ and $\hat{Q}_{ij} = \exp(\mathbf{w}_i^\top \tilde{\mathbf{w}}_j)$.
- Squared error on logs: $(\log \hat{Q} - \log \hat{P})^2$.
- The weight X_i is replaced by the bounded $f(X_{ij})$.

The result is exactly the GloVe objective. The two methods use the same statistics with different losses.

Where GloVe sits

	LSA / PPMI + SVD	**skip-gram**	**GloVe**
statistics	global counts	local windows	global counts
target	counts / PMI	next-context prob.	$\log X_{ij}$
loss	unweighted squared (SVD)	cross-entropy (sampled)	weighted squared
weighting	uniform over all cells	implicit $\propto X_i$	explicit, capped $f(X_{ij})$
cost per epoch	one-off SVD	$O(\lvert C \rvert)$	$O(\mathrm{nnz}(X))$
linear analogies	weak	strong	strong



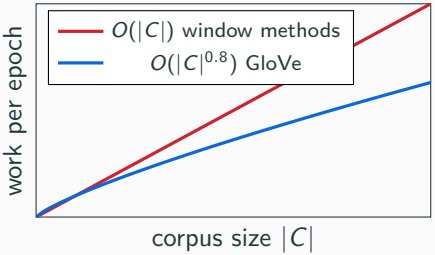
Complexity

How many non-zeros? A power-law argument

Model the co-occurrence counts as a power law in their frequency rank r: $X_{ij} = k/r_{ij}^{\alpha}$. Since the corpus size is $|C| \propto \sum X_{ij}$, a generalised harmonic sum gives

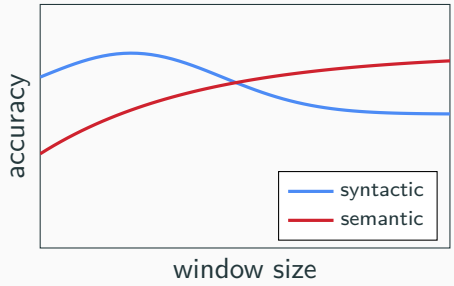
$$|C| \sim k\,\frac{|X|^{1-\alpha}}{1-\alpha} + k\,\zeta(\alpha) \quad \Longrightarrow \quad |X| = \begin{cases} O(|C|) & \alpha < 1 \\ O\big(|C|^{1/\alpha}\big) & \alpha > 1 \end{cases}$$

- The paper fits $\alpha \approx 1.25$, so $|X| = O(|C|^{0.8})$.
- GloVe's cost per epoch is $O(|X|)$, which is sub-linear in corpus size.
- Window-based methods scale as $O(|C|)$.



Behaviour & limitations

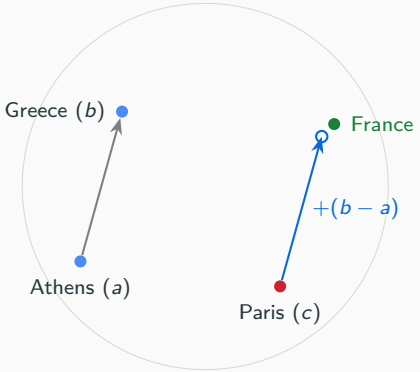
Effect of the main choices (qualitative)



schematic, after Pennington et al. (2014), Fig. 2

- **Window size**: small windows capture *syntax* (local, word-order cues); large windows capture *semantics* (topical association).
- **Dimension** d: gains flatten out around 200–300.
- **Corpus size**: more data steadily helps semantic tasks, and the corpus domain matters (encyclopaedic text helps factual analogies).
- $x_{\max}$, α: fairly robust; $\alpha = \frac{3}{4}$ beats $\alpha = 1$.

Evaluating embeddings



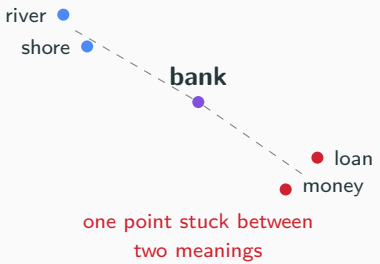
Analogy ($a : b :: c :?$): pick the word closest in cosine to $\mathbf{w}_b - \mathbf{w}_a + \mathbf{w}_c$, excluding a, b, c. Split into semantic (capitals, family, ...) and syntactic (tense, plural, ...) questions.

Similarity: rank correlation between $\cos(\mathbf{w}_i, \mathbf{w}_j)$ and human ratings (WordSim-353, SimLex, ...).

Downstream: use the vectors as features, e.g. for named-entity recognition.

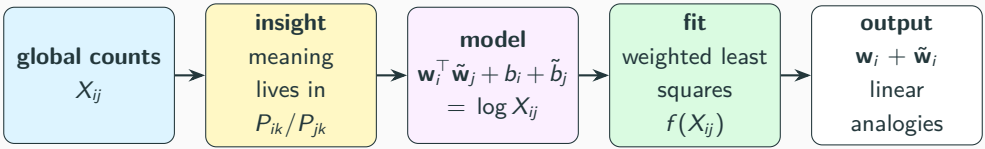
Limitations

- **One vector per word type**: senses are blended (*bank*: river vs. finance).
- **Static**: the same vector in every sentence. Contextual models such as ELMo (Embeddings from Language Models) and BERT (Bidirectional Encoder Representations from Transformers) address this.
- **Closed vocabulary**: no vector for unseen words, and no subword information (unlike fastText).
- **Inherits corpus biases**, e.g. gender and occupation associations.
- **Symmetric context** loses word order beyond window distance.



Summary

GloVe in one picture



It combines the statistical efficiency of global matrix methods with the linear structure of prediction methods.
Equivalently, it is a weighted factorisation of a shifted PMI matrix with learned marginals.

References

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